

Curriculum Map

Course Title: General Science

Grade: 6



Department of Biology

Examination paper for BI2034 Community Ecology and Ecosystems

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Examination date: 29th May 2017

Examination time (from-to): 0900-1300

Permitted examination support material: A basic calculator is allowed.

Other information: Answer all questions

Language: English/Bokmål

Students will find the examination results in Studentweb. Please contact the department if you have questions about your results. The Examinations Office will not be able to answer this.

Number of pages (front page excluded): 2

Number of pages enclosed: 3

Essential Question(s): Unit EQ: How can we use our natural resources in a more responsible way?	
Unit (Name/Number): Natural Resources	Pacing: Non-Renewable Resources (10 days)
Lesson EQs: What are the forms and origin of nonrenewable resources currently in use?	
king av eksamensoppgave Originalen er:	Renewable Resources (5 days)
How do we use nonrenewable resources?	Environmental History (3 days)
1-sidig ☐ 2-sidig ☐	Resources in Pennsylvania (4 days)
What are the benefits and drawbacks of using nonrenewable resources?	
What are some sources and uses of renewable resources?	
sort/hvit ☐ farger ☐	
What are the advantages and disadvantages of using renewable resources?	
What natural resource life figures have shaped the environmental history of the United States?	
What types of renewable and nonrenewable resources are available in PA and how are they managed?	

<u>Content/Key Concepts</u>	<u>Standards</u>	<u>Key Vocabulary</u>	<u>Learning Activities/Resources</u>	<u>Evidence of Learning</u> (Assessments; Performance Tasks)
<u>Non-Renewable Resources</u>	<u>3.2.6.A4</u> <u>3.2.6.B1</u> <u>3.2.6.B2</u> <u>3.2.7.B2</u> <u>3.3.8.A2</u> <u>3.3.8.A3</u>	• <u>Non-renewable</u> • <u>natural resource</u> • <u>fossil fuel</u> • <u>oil (petroleum)</u> • <u>natural gas,</u> • <u>coal (peat, lignite, bituminous and anthracite)</u> • <u>hydrocarbons</u> • <u>combustion</u> • <u>fuel</u>	• <u>Environmental Science Textbook</u> • <u>PowerPoints</u> • <u>Videos</u> • <u>Quizlet (Vocab)</u> • <u>Steam Engine Demonstration</u> • <u>Summarizer</u>	<u>Quiz</u> <u>Unit Test (with Renewable Resources)</u>

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	4.5.6.A 4.5.7.C 4.5.7.D 4.5.7.E	• nuclear energy • Uranium-235 • Fission • Fusion • Methane • global warming • greenhouse effect • acid rain		
Renewable Resources	3.2.6.A4 3.2.6.B1 3.2.7.B2 4.3.7.A 4.3.8.A 4.5.7.A	• renewable resource • biomass • biomass • ethanol • geothermal • solar energy • wind power • hydroelectric energy • tidal/wave energy	• Environmental Science Textbook • PowerPoints • Videos • Lab • Quizlet (Vocab) • Summarizer	Quiz Unit Test (with Non-Renewable Resources)
Environmental History	4.5.6.A 4.5.6.C 4.5.7.A	• Rachel Carson (1962) • Gifford Pinchot (1907) • Theodore Roosevelt (1903) • John Muir (1892) • Deep Water Horizon (2010) • Clean Water Act (1972) • Clean Air Act (1963) • Three Mile Island (1979) • Exxon Valdez (1989) • EPA (1970)	• Videos • PowerPoints • Environmental Science Textbook	Quiz/Timeline Project

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PA Resources and Their Management	4.3.7.B 4.5.8.C 4.5.6.A 4.5.6.D	• Conservation • Preservation • Exploitation • Sustainability • waste management • recycling • landfill • Incineration • hazardous wastes • toxic • corrosive • Radioactive • Flammable • explosive • Waste stream	• Videos • PowerPoints • Environmental Science Textbook • EPA and DCNR websites • Lab	Test on Environmental History and Resources in PA
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Checked by:

Date _____ Signature _____

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English

Task 1.

- a) ~~Discuss how diversity can facilitate stability in ecological communities. Include descriptions of the redundancy hypothesis, insurance hypothesis and portfolio hypothesis.~~
- b) ~~What evidence exists for diversity-stability relationships in ecology?~~
- c) ~~How is community invasibility related to diversity? Describe patterns and mechanisms.~~

Task 2.

- a) ~~Describe the resource-ratio hypothesis in the context of succession. Illustrate this with one or more figures.~~
- b) ~~Explain the difference between founder and dominance control in ecological communities, and give examples of contexts where each process is prevalent.~~
- c) ~~Discuss how lottery models can be applied to understand successional processes in communities. In what ways do lottery models differ from the species-sorting perspective?~~

Task 3.

- a) ~~Explain what you understand with a "species-accumulation-curve".~~
- b) ~~Explain what it means that a society has high or low equitability (the term "evenness" applies the same), and explain how equitability can be measured.~~
- c) ~~A biologist conducted a community ecological study and collected standardized insect samples in two different habitats. In the first habitat, the biologist collected 30 samples and recorded 30 species and the habitat had high equitability ("evenness"). In the second habitat, the~~

biologist collected 100 samples and also here he found 30 insect species, but equitability was significantly lower compared to habitat. 1.
Create a "Rank abundance diagram" where you draw simple sketches of the "Rank abundance curves"» For habitats 1 and 2, respectively,
based on the information provided. Explain why the two curves have different shapes.

Bokmål

Oppgave 1.

- a) Forklar hvordan diversitet kan fasilitere stabilitet i økologiske samfunn. Inkluder beskrivelser av hypotesen om overflodighet (engelsk: «redundancy hypothesis»), forsikringshypotesen («insurance hypothesis») og portefoliehypotesen («portfolio hypothesis»).
- b) Hvilke bevis finnes for diversitet-stabilitet-forhold i økologi?
- c) Hvordan er diversitet relatert til evnen biologiske samfunn har til å motstå spredning og etablering av fremmede arter? Beskriv mekanismer og observerte mønstre.

Oppgave 2.

- a) Beskriv ressursforholdshypotesen (en: «the resource-ratio hypothesis») i sammenheng med suksesjon. Illustrer med en eller flere figurer.
- b) Beskriv forskjellen mellom grunnlegger- og dominanskontroll i økologiske samfunn, og gi eksempler på tilfeller hvor hver prosess er dominant.
- c) Diskuter hvordan lotterimodeller kan brukes for å forstå suksesjonsmønstre i økologiske samfunn. Hvordan skiller lotterimodeller seg fra et perspektiv som går på sortering av arter, eller «artsorteringsperspektivet».

Oppgave 3.

- a) Forklar hva du forstår med en «arts-akkumulerings-kurve» («species accumulation curve»).
- b) Forklar hva det betyr at et samfunn har høy eller lav «equitability» (begrepet «evenness» brukes om det samme), og forklar hvordan equitability kan måles.

c) ~~En biolog gjennomførte et samfunnsøkologisk studie og samlet inn standardiserte insektprøver i to ulike habitat. I det første habitatet samlet biologen inn 30 prøver og registrerte der 30 insektarter og habitatet hadde høy equitability («evenness»). I det andre habitatet samlet biologen inn 100 prøver og fant også her 30 insektarter, men equitability var her betydelig lavere enn i habitat 1. Lag et enkelt «Rank—abundance diagram» der du tegner inn enkle skisser av «Rank—abundance kurvene» for habitat 1 og 2 basert på informasjonen som er gitt. Forklar hvorfor de to kurvene har ulik form.~~